

**SIMATS ENGINEERING**  
**ASSESSMENT TOOL 1 - Low (Pseudo-code Writing)**

**COURSE CODE: CSA0910**

**COURSE NAME: Programming in Java**

**COURSE OUTCOME COVERED**

**CO1:** *Apply object-oriented programming principles such as classes, objects, encapsulation, data hiding, inheritance, and polymorphism to design and implement entity manipulations (BL3,BL4)*

**QUESTIONS: 10**

**Total Marks: 100**

**ANNEXURE A PSEUDO-  
CODE WRITING**

***Code of Conduct:***

*I S.ROHINI / 192424413 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

***Signature of the student: S.Rohini***

**RUBRICS FOR CALCULATION:**

| <b>Criteria</b>           | <b>Weightage</b> | <b>Level 4 – Exemplary</b>                                            | <b>Level 3 – Proficient</b>               | <b>Level 2 – Developing</b>  | <b>Level 1 – Beginning</b>      |
|---------------------------|------------------|-----------------------------------------------------------------------|-------------------------------------------|------------------------------|---------------------------------|
| Problem Understanding     | 20%              | Identifies all inputs, outputs, constraints accurately                | Minor missing elements                    | Partial understanding        | Incorrect interpretation        |
| Algorithm Design          | 30%              | Complete, correct, and logically sound solution                       | Minor logical gaps                        | Partially correct logic      | Incorrect approach              |
| Use of Control Structures | 20%              | Efficient and appropriate use of loops, conditions, functions/modules | Minor inefficiencies                      | Limited or basic usage       | Incorrect or missing constructs |
| Pseudocode Structure      | 10%              | Well-structured, clear, consistent format, easy to follow             | Mostly clear with minor issues            | Difficult to follow in parts | Poorly structured and unclear   |
| Completeness              | 20%              | Handles all cases; optimized approach                                 | Handles most cases; reasonable efficiency | Handles basic cases only     | Incomplete or inefficient       |

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CSA 0910 → JAVA

## C01 - AT3 → Pseudocode Writing

### 1) STUDENT MANAGEMENT SYSTEM:

START

CLASS Student

PRIVATE : name, rollNo, marks

METHOD acceptDetails()

INPUT name, rollNo, marks

END

METHOD DisplayResults()

IF marks  $\geq$  50

PRINT "PASS"

ELSE

PRINT "FAIL"

END

END CLASS

MAIN

CREATE Students

CALLS. acceptDetails()

CALL s. displayResults()

STOP

### 2) LIBRARY BOOK MANAGEMENT:

START

CLASS Book

PRIVATE : title, author, ISBN, isAvailable

METHOD IssueBook()

isAvailable = False

END

METHOD returnBook()

```

END
METHOD Display ()
    PRINT title, author, ISBN, ISAvailable
END
END CLASS

MAIN
CREATE b1, b2, b3
CALL b1.issuebook ()
CALL b1.returnbook ()
CALL b1.Display ()
STOP

```

### 3) EMPLOYEE SALARY CALCULATION

```

START
CLASS Employee
    PRIVATE : empID, name, basicpay
    METHOD calculateSalary ()
        gross = basicpay + HRA + DA
        net = gross - PF - TAX
    END
    METHOD Display ()
        PRINT empID, name, gross, net
    END
END CLASS

MAIN
CREATE Employee e
INPUT details
CALL e.calculateSalary ()
CALL e.Display ()
STOP

```



#### 4) SHAPE AREA CALCULATOR:

START

CLASS shape

METHOD calculateArea()

END

CLASS Circle EXTENDS shape

OVERRIDE calculateArea()

END

CLASS RECTANGLE EXTENDS shape

OVERRIDE calculateArea()

END

CLASS Triangle EXTENDS shape

OVERRIDE calculateArea()

END

MAIN

CREATE Circle, Rectangle, Triangle

CALL calculateArea()

STOP

#### 5) VEHICLE REGISTRATION SYSTEM:

START

Vehicle = displayDetails()

(Two Wheeler

Four Wheeler

MAIN

CREATE objects

CALL displayDetails()

STOP

## 6) BANK ACCOUNT HIERARCHY

```
START
BankAccount → withdraw()
SavingsAmount
CurrentAmount
MAIN
CREATE Objects
CALL withdraw()
STOP
```

## 7) HOSPITAL PATIENT RECORD SYSTEM:

```
START
CLASS Patient
    PRIVATE diagnosis, prescription
    METHOD generateBill()
END
CLASS Inpatient EXTENDS Patient
    OVERRIDE generateBill()
END
MAIN
CREATE Inpatient()
CREATE Outpatient
CALL generateBill()
STOP
```

## 8) E-COMMERCE PRODUCT CATALOG:

```
START
product → applyDiscount()
Electronics
clothing
```



Grocery

MAIN

CREATE objects

CALL ApplyDiscount()

STOP

9) UNIVERSITY STAFF HIERARCHY:

START

Staff

Teaching Staff

professor

Lecturer

Non teaching Staff

MAIN

CALL calculate Allowance()

STOP

10) SMART HOME DEVICE CONTROLLER:

START

Smart Device

execute Command()

Smartlight

Override executeCommand()

Smart thermostat

Override execute Command()

Smart lock

Override execute Command()

Hide PIN

MAIN

Create all objects

CALL executeCommand("ON")

STOP

